THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** independent variable	The quantity changed by the investigator is the independent variable.
2	**(c)** Robert Hooke	Hooke saw box-like compartments and called them cells.
3	**(d)** physical, mental and social well-being	Health is more than the absence of disease.
4	**(a)** cell or battery	A cell converts chemical energy to electrical energy.
5	**(a)** a question or curiosity about an observation	Questions drive investigations.
6	**(d)** cell, tissue, organ, organ system, organism	Cells form tissues; tissues form organs, and so on.
7	**(c)** something a doctor can observe or measure, such as a rash	Signs are observable.
8	**(d)** nichrome	Nichrome has high resistance and a high melting point.
9	**(a)** length of the pendulum	Length is the quantity that is changed.
10	**(b)** vacuole	A large vacuole stores water and substances.
11	**(a)** vitamin C	Bleeding gums and weak joints.
12	**(a)** a resistor	Resistor symbol.
13	**(b)** dependent variable	It is the quantity that is measured as the result.
14	**(a)** microscope	A microscope magnifies tiny things.
15	**(a)** getting along well with family and neighbours	Social health concerns relationships.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Testable explanation or guess before an experiment (1); e.g. a puri puffs more in hotter oil (1).	1+1
17	Robert Hooke in 1665 (1); He observed thin slices of cork under a microscope (1).	1+1
18	State of complete physical, mental and social well-being (1); Not merely the absence of disease (1).	1+1
19	A closed path along which electric current flows (1); Includes a source, wires and a device (1).	1+1
20	To check that the result is not a chance event (1); To get a more dependable (average) value (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Question and hypothesis: hotter oil makes puffing faster (1); Vary temperature; keep dough, thickness and size same (1); Measure time with stopwatch, repeat, record in a table (1).	1+1+1
22	Peel a thin layer and place it in water on a slide (1); Add a drop of stain and cover with a cover slip carefully (1); Observe under the microscope and draw the rectangular cells (1).	1+1+1
23	Physical: body works well, e.g. energy for play (1); Mental: emotional balance, e.g. managing stress (1); Social: good relationships, e.g. helping neighbours (1).	1+1+1
24	Cell, switch and bulb connected in a loop with wires (1); When the switch is closed the path is complete (1); Current flows through the filament and the bulb glows (1).	1+1+1
	OR: Real work often moves back and forth (1); A surprising result may send scientists back to change the question (1); So the steps are repeated and revised (1).	1+1+1
25	All living bodies are made of cells (1); Each cell performs life functions (1); New cells arise from existing cells (1).	1+1+1
	OR: Vitamin A: night blindness (1); Vitamin C: scurvy, bleeding gums (1); Iron: anaemia, weakness (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Question stated clearly (1); Hypothesis and independent/dependent variables (1); Constants: oil temperature, thickness, size, frying time (1); Procedure: vary water, fry, observe puffing/time (1); Repeat and record in a table (1).	1+1+1+1+1

	OR: Cell wall: shape and protection (1); Cell membrane: controls entry/exit (1); Cytoplasm: site of cell activities (1); Nucleus: controls cell and has genetic material (1); Chloroplast and large vacuole: food making; storage (1).	1+1+1+1+1
27	WHO: complete well-being, not just absence of disease (1); Physical: normal body functions (1); Mental: positive emotions, coping with stress (1); Social: healthy relationships (1); All aspects are linked (1).	1+1+1+1+1
	OR: Set-up: wire, cell, switch, compass (1); Needle deflects when current flows (1); Conclusion: current has a magnetic effect (1); Reverse current: deflection reverses (1); Needle returns when current is off (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** a gap in the circuit	An open switch breaks the path.
2	**(b)** a flexible approach that may be repeated and revised	Real investigations loop back and revise earlier steps.
3	**(c)** tissue	A tissue is a group of similar cells.
4	**(c)** chronic	Chronic diseases are long lasting.
5	**(a)** low melting point	It melts and breaks the circuit during overload.
6	**(b)** check the method and repeat the experiment	Differences must be investigated.
7	**(b)** give shape and protection to plant cells	It is rigid and strong.
8	**(a)** vitamin A	Needed for vision.
9	**(b)** a broken connection or a fused filament	Open circuit.
10	**(c)** thickness of the rolled dough	All factors other than the one being studied are kept constant.
11	**(b)** nucleus	The nucleus holds the genetic material and directs cell activities.
12	**(a)** communicable	Communicable diseases are infectious.
13	**(b)** from the positive terminal through the circuit to the negative terminal of the cell	Conventional current is from the positive terminal.
14	**(b)** revise the hypothesis and test again	Honest reporting is essential.
15	**(a)** an organ	It is made up of several tissues working together.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	It contains the genetic material (DNA) (1); It controls all activities of the cell (1).	1+1
17	They prepare the body to fight a particular pathogen (1); They prevent the disease before it occurs (1).	1+1
18	It has a very high melting point (1); It can glow without melting (1).	1+1
19	Keep long hair and loose clothing tied back (1); Use tongs or holders; never leave the flame unattended (1).	1+1
20	Bacteria lack a membrane-bound nucleus (1); Their genetic material lies free in the cytoplasm (nucleoid) (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	It is connected in series (1); Excess current heats and melts it (1); The circuit breaks and the appliances are safe (1).	1+1+1
22	A test in which only one factor is changed (1); Others are kept the same (1); Otherwise we cannot tell which factor caused the effect (1).	1+1+1
23	Cell wall is outside, rigid, in plants only (1); Cell membrane is thin and flexible, in all cells (1); Membrane controls what enters and leaves the cell (1).	1+1+1
24	They work only on bacteria (1); Wrong use causes resistance in bacteria (1); May have side effects (1).	1+1+1
	OR: Electric iron (1); Geyser, toaster, heater (1); Bulb filament (1).	1+1+1
25	Temperature of the oil (1); Amount and kind of flour and water in the dough (1); Size of the puri and time kept in oil (1).	1+1+1
	OR: Plant cell has a cell wall; animal cell does not (1); Plant cell has chloroplasts; animal cell does not (1); Plant cells have a large vacuole; animal cells have small ones (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Parallax or wrong reading of scale (1); Faulty or uncalibrated instrument (1); Repeat the readings (1); Take the average (1); Use proper instrument and procedure (1).	1+1+1+1+1

	OR: Bacteria: single-celled, no nucleus (1); Viruses: non-cellular, parasites (1); Fungi: lack chlorophyll; yeast, mushroom (1); Protozoa: single-celled, with nucleus; Amoeba (1); Each can cause disease in some cases (1).	1+1+1+1+1
27	Personal: handwashing, bathing, clean clothes (1); Community: clean water (1); Community: waste disposal (1); Community: drainage and sanitation (1); Together they break the chain of infection (1).	1+1+1+1+1
	OR: Crane: current ON lifts scrap (1); Current OFF drops it (1); Magnet separates iron from rubbish (1); Soft iron is easily magnetised and demagnetised (1); Steel would stay magnetised (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** tuberculosis	TB can last for months.
2	**(b)** tungsten	Tungsten has a very high melting point.
3	**(c)** curiosity and willingness to question	Science asks for evidence.
4	**(c)** a simple microscope with a single lens	He ground his own lenses.
5	**(d)** anaemia	Low haemoglobin.
6	**(c)** a thin nichrome wire	High resistance means more heating.
7	**(b)** hypothesis	A hypothesis is a testable guess.
8	**(a)** cell membrane	It is selectively permeable.
9	**(d)** diabetes	It does not spread from person to person.
10	**(a)** copper	Metals conduct electricity.
11	**(b)** organise data and see patterns	Tables show trends clearly.
12	**(a)** multicellular	Many cells make the body.
13	**(b)** disease-causing microorganisms	Pathogens cause diseases.
14	**(a)** magnetic effect of current	An electromagnet pulls the hammer.
15	**(a)** away from you and others	Hot liquid may spurt out.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Acute: short duration, e.g. cold (1); Chronic: long lasting, e.g. diabetes (1).	1+1
17	High resistance gives more heat (1); High melting point and does not oxidise easily (1).	1+1
18	A factor or quantity that can change (1); Independent (changed) and dependent (measured) (1).	1+1
19	Group of similar cells doing a particular function (1); e.g. muscle tissue or xylem (1).	1+1
20	Symptom is felt by the patient, e.g. pain (1); Sign is observed or measured, e.g. fever or rash (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Air: cold, tuberculosis (1); Water and food: cholera, typhoid (1); Vectors or contact: malaria, ringworm (1).	1+1+1
22	Open switch makes a gap (1); The circuit is incomplete (1); No current flows (1).	1+1+1
23	Curiosity and asking questions (1); Honesty in recording and reporting data (1); Patience and willingness to repeat and revise (1).	1+1+1
24	Yeast: bread making (1); Bacteria: curd formation from milk (1); Fungi (e.g. moulds): production of antibiotics like penicillin (1).	1+1+1
	OR: Washing hands with soap (1); Brushing teeth and bathing daily (1); Wearing clean clothes and trimming nails (1).	1+1+1
25	Do not touch bare wires (1); Do not use wet hands on switches (1); Do not overload sockets (1).	1+1+1
	OR: Everyone gets comparable values (1); Results can be repeated and checked by others (1); Using body parts or guesswork gives different values (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Observe and ask a question (1); Form a hypothesis (1); Plan and perform a fair test (1); Analyse data and conclude (1); New questions arise, so steps repeat (1).	1+1+1+1+1
	OR: Cell: e.g. muscle cell (1); Tissue: group of muscle cells (1); Organ: heart (1); Organ system: circulatory system (1); Organism: the whole body (1).	1+1+1+1+1

27	Malaria: Anopheles carries Plasmodium (1); Dengue: Aedes mosquito carries a virus (1); Cholera: contaminated water (1); Prevention: avoid stagnant water, nets (1); Boil water, hygiene (1).	1+1+1+1+1
	OR: Wind insulated wire around an iron nail (1); Connect to a cell (1); Test by lifting paper clips (1); More turns means more clips (1); More current means stronger (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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